

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
Screening Test - Kaprekar Contest

(Maths Olympiad at **SUB-JUNIOR LEVEL** VII to VIII Standards)
Saturday, 4th September 2004.

Note:

- 1) Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.
- 2) Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- 3) For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.
- 4) Diagrams are only visual aids; they are not drawn to scale.
- 5) You are free to do rough work on separate sheets.
- 6) Duration of the test: 2 p.m. to 4. p.m.

1. A four digit number of the form $abaa$ (a 's and b 's are the digits of the four digit number) is divisible by 33. The number of such four digit numbers is

(A) 36 (B) 6 (C) 3 (D) 1

2. If $a + 2a + 3a + \dots + 1000a = 2b + 4b + 6b + \dots + 2000b$
 $= 3c + 6c + 9c + \dots + 3000c$

then $a:b:c$ is as

(A) 1:2:3 (B) 3:2:1 (C) 2:3:6 (D) 6:3:2

3. If $a^* = a + 1$ and $*a = a - 1$, then

$1^* - *1 + 2^* - *2 + 3^* - *3 + \dots + 1000^* - *1000$ is equal to

(A) 1000 (B) -1000 (C) 2000 (D) -2000

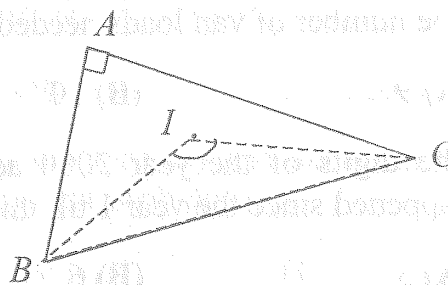
4. ' a ' and ' b ' are two natural numbers with $a + b = 8$. If $a \geq b$ and $a^2 + b^2$ has minimum value, then a and b are given by

(A) 7,1 (B) 6,2 (C) 4,4 (D) 5,2

5. In the adjoining figure, $\triangle ABC$ is right angled at A ; $m\angle B = m\angle C$.

The bisectors of angle B and C meet at I . Then $m\angle BIC$ is

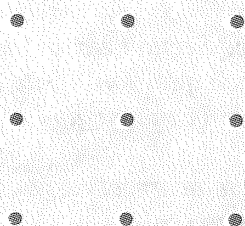
(A) 135° (B) 115° (C) 100° (D) 90°



6. The number of isosceles triangles in which one angle is 4 times another angle is

(A) 2 (B) 1 (C) Infinitely many (D) 4

7.



Nine dots are arranged such that they are equally spaced horizontally and vertically as in the figure.

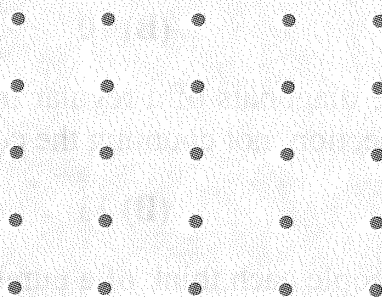
The number of triangles which are not right angled triangles that can be formed with the above dots as vertices is

- (A) 18 (B) 21 (C) 40 (D) 24
8. The number of non negative integers which are less than 1000 and end with only one zero is
- (A) 90 (B) 99 (C) 91 (D) 100
9. The tenth term of the sequence (2,5), (3,7), (5,11), (7,13), (11,17), is
- (A) (23,29) (B) (19,13) (C) (20,04) (D) (29,31)
10. The sum of the digits of the number $10^n - 1$ is 3798. The value of n is
- (A) 431 (B) 673 (C) 422 (D) 501
11. The image of INMO when reflected in a mirror is
- (A) IWVO (B) OMNI (C) INWO (D) OWNi
12. A transport company's vans each carry a maximum load of 12 tonnes. 24 sealed boxes each weighing 5 tonnes have to be transported to a factory. The number of van loads needed to do this is
- (A) 9 (B) 10 (C) 11 (D) 12
13. The digits of the year 2000 add up to 2. In how many years has this happened since the year 1 till this year 2004?
- (A) 3 (B) 6 (C) 9 (D) 10

14. Ram is 7 years younger than Ravi. In four years time, Ram will be half of Ravi's age. The sum of their age now is

- (A) 13 (B) 15 (C) 17 (D) 19

15. A circle is added to the equally spaced grid alongside. The largest number of dots that the circle can pass through is



- (A) 4 (B) 6 (C) 18 (D) 10

16. A certain number has exactly eight factors including 1 and itself. Two of its factors are 21 and 35. The number is

- (A) 105 (B) 210 (C) 420 (D) 525

17.

13		
5		15
x		

In a magic square, each row and each column and both main diagonals have the same total. The number that should replace x in this partially completed magic square is

- (A) more information needed (B) 9
(C) 10 (D) 12

18. In the triangle ABC , D is a point on the line segment BC such that $AD = BD = CD$. The measure of angle BAC is

- (A) 60° (B) 75° (C) 190° (D) 120°

19. The product of Hari's age in years on his last birthday and his age now in complete months is 1800. Hari's age on his last birth day was

- (A) 9 (B) 10 (C) 12 (D) 15

20. One hundred and twenty students take an exam which is marked out of 100 (with no fractional marks). No three students are awarded the same mark. What is the smallest possible number of pairs of students who are awarded the same mark?

- (A) 9 (B) 10 (C) 19 (D) 20

21. If all the diagonals of a regular hexagon are drawn, the number of points of intersection, not counting the corners of the hexagon is

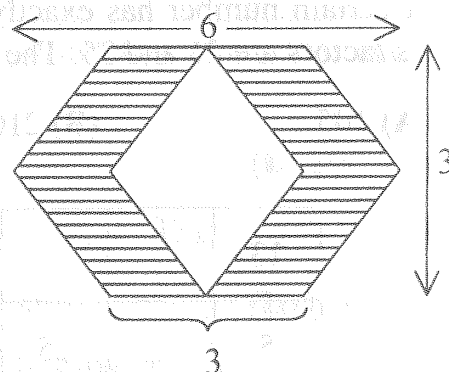
- (A) 6 (B) 13 (C) 7 (D) 12

22. Three people each think of a number, which is the product of two different primes. The product of the three numbers which are thought of is

- (A) 120 (B) 12100 (C) 240 (D) 3000

23. The area of the shaded region in the diagram is

- (A) 9 (B) $3\sqrt{2}$
(C) 18 (D) $6\sqrt{3} - 3\sqrt{2}$



24. The largest positive integer which cannot be written in the form $5m + 7n$ where m and n are positive integers is

- (A) 25 (B) 35
(C) great then 100 (D) great than 350

25. The last digit in the finite decimal representation of the number $\left(\frac{1}{5}\right)^{2004}$ is

- (A) 2 (B) 4 (C) 6 (D) 8